

HW 6

cyclic prefix

1. Consider std. LTE CP duration to be $4.7 \mu\text{s}$. If shortest path between TX & RX 1km, what's the dist traveled by longest path to ensure its delays remain w/in $4.7 \mu\text{s}$ (CP duration). (signals prop @ $c = 3 \times 10^8 \text{ m/s}$)

$$\overset{1\text{K}}{\downarrow} \\ 1000 + 3 \times 10^8 (4.7 \times 10^{-6}) = \boxed{2410 \text{ m}}$$

2. Two subcarriers $\cos(2\pi f_1 t)$ & $\cos(2\pi f_2 t)$ are orthogonal over interval $[0, T_s]$ if $\int_0^{T_s} \cos(2\pi f_1 t) \cos(2\pi f_2 t) dt = 0$ show that
9.

$$\text{i. } f_1 + f_2 = n/(2T_s), n \text{ integer}$$

$$\text{ii. } f_1 - f_2 = k/(2T_s), k \text{ integer}$$

to satisfy the ortho cond.

$$\cos(2\pi f_1 t) \cos(2\pi f_2 t) = \frac{1}{2} (\cos(2\pi t(f_1 - f_2)) + \cos(2\pi t(f_1 + f_2)))$$

$$\frac{1}{2} \int_0^{T_s} \cos(2\pi t(f_1 - f_2)) + \cos(2\pi t(f_1 + f_2)) dt$$

$$\frac{1}{2} \left[\int_0^{T_s} \frac{\sin(2\pi t(f_1 - f_2))}{2\pi(f_1 - f_2)} + \frac{\sin(2\pi t(f_1 + f_2))}{2\pi(f_1 + f_2)} \right]$$

$$\frac{1}{2} \left(\frac{\sin(2\pi T_s(f_1 - f_2))}{2\pi(f_1 - f_2)} + \frac{\sin(2\pi T_s(f_1 + f_2))}{2\pi(f_1 + f_2)} \right)$$

passing in $n/(2T_s)$ for $f_1 + f_2$ and $\frac{k}{(2T_s)}$ for $f_1 - f_2$

$$\frac{1}{2} \left(\frac{\sin(k\pi)}{k\pi/T_s} + \frac{\sin(n\pi)}{n\pi/T_s} \right)$$

B/c $\sin$ of $\pi \times$ any int will always $= 0$, this shows that $f_1 + f_2$ and $f_1 - f_2$ must equal some integer each $\times \frac{1}{2T_s}$ to satisfy orthogonality.

b. min freq spacing possible to maintain ortho?

well it's given isn't it?

$f_1 - f_2 = k/(2T_s)$ so have $k=1$ (0 would just mean no transfer)

$$\boxed{f_1 - f_2 = 1/(2T_s)}$$

3. The actual BW occupied in std. 20 MHz LTE is only 18 MHz.
Symbol duration is $T_s = 66.667 \mu s$

a. sub carrier spacing to nearest Hz $= \frac{1}{T_s} = \frac{1}{66.667 \mu s} = \boxed{15 \text{ kHz}}$

b. How many sub carriers to span BW:

$$\frac{18 \text{ M}}{15 \text{ k}} = \boxed{1200 \text{ sub carriers}}$$

c. Nearest power-of-two IFFT required for OFDM signal

$2^{10} = 1024 < 1200 < 2048 = 2^{11}$ 2¹¹ or 2048 IFFT required for all sub carriers

4. Consider OFDM parameters

- carrier freq $f_c = 2.6 \text{ GHz}$
- Max velo of mobile terminal $v = 135 \text{ km/h}$
- max delay spread $\geq 5 \text{ ms}$
- QPSK modulation w/ net data rate $R = 4 \text{ Mbit/s}$

9. min duration of the CP (denoted by T_g) to completely avoid ISI (inter symbol interference)?

$$T_g \geq \text{max delay spread} \geq 5 \text{ ms}$$

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b. Find Doppler spread. (max doppler freq corresponds to $v = 135 \text{ km/h}$ at $f_c = 2.6 \text{ GHz}$)

$$f_d = f_c \frac{v}{c} = 2 \times 10^9 \left(\frac{135 \times 1000}{3600 \times 3 \times 10^8} \right) = 250 \text{ Hz}$$

c. Find coherence time (T_c) of channel (inverse of doppler spread)

$$\frac{1}{f_d} = \frac{1}{250} = 4 \text{ ms} \geq T_c$$

d. Assuming the actual symbol interval T_s is 2% of T_c , find sub-carrier spacing Δf , the required # of sub-carriers N , and the system BW (B) for support the net data rate $R = 4 \text{ Mbit/s}$. Note: $R = N k / (T_s + T_g)$
 $k = \text{info bits carried per symbol}$ (2 for QPSK)

$$T_c \times 0.02 = 8 \times 10^{-5} = T_B$$

$$4 \times 10^6 \text{ bits/s} = N(2) / (8 \times 10^{-5} + 5 \times 10^{-6})$$

$$N = 170 \text{ sub-carriers}$$

$$\Delta f = \frac{1}{T_B} = 12.5 \text{ kHz}$$

$$B = N \cdot \Delta f = 2.125 \text{ MHz}$$